

TORPABRON, Sweden

WMO: 025150

Lat: 58.1997N

Lon: 12.1541E

Elev: 17

StdP: 101.12

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-13.3	-18.2	0.8	-16.5	-14.8	1.0	-12.7	14.1	7.9	12.3	6.8	0.4	90	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	27.2	18.7	25.3	18.1	23.2	17.0	20.0	25.0	19.0	23.3	18.0	21.9	3.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	13.2	21.5	17.3	12.4	20.6	16.5	11.7	19.7	57.3	25.3	54.0	23.6	50.8	21.9	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.0	7.8	-20.4	29.8	5.4	1.6	-24.3	30.9	-27.5	31.9	-30.5	32.8	-34.4	33.9		
(5)			-20.7	21.2	4.7	1.3	-24.1	22.2	-26.9	22.9	-29.5	23.6	-32.9	24.6	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-1.9	-1.1	1.6	6.3	10.8	14.3	17.2	16.3	12.2	7.5	4.1	0.2	(6)	
	DBStd	7.61	5.78	4.79	4.22	3.11	3.06	2.55	2.40	2.43	2.71	3.81	4.09	5.68	(7)	
	HDD10.0	1671	368	309	259	117	25	1	0	0	11	95	181	305	(8)	
	HDD18.3	4040	626	543	518	361	233	124	53	71	184	336	428	563	(9)	
	CDD10.0	701	0	0	0	6	51	129	222	197	77	17	2	0	(10)	
	CDD18.3	28	0	0	0	0	0	3	17	8	0	0	0	0	(11)	
	CDH23.3	386	0	0	0	0	9	81	209	85	2	0	0	0	(12)	
	CDH26.7	62	0	0	0	0	1	13	40	8	0	0	0	0	(13)	
(14) Wind	WSAvg	3.2	3.5	3.3	3.4	3.2	3.6	3.2	2.9	2.8	3.4	2.9	3.4	3.3	(14)	
(15) Precipitation	PrecAvg	946	87	66	56	56	56	80	84	88	91	103	90	89	(15)	
	PrecMax	1241	171	163	122	123	103	163	176	178	217	209	177	206	(16)	
	PrecMin	675	9	19	13	10	11	8	6	7	20	27	29	16	(17)	
	PrecStd	150	46	39	27	32	24	35	45	39	46	46	44	52	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	9.1	14.7	20.0	24.4	28.6	30.1	27.7	23.4	17.3	13.6	10.8	(19)	
		MCWB	7.4	7.8	8.5	12.1	15.1	17.9	20.4	19.4	17.1	13.3	12.4	9.8	(20)	
	2%	DB	7.4	7.3	11.2	17.6	21.9	25.6	28.0	25.8	20.5	15.2	12.1	9.5	(21)	
		MCWB	6.2	5.8	6.8	10.5	14.3	16.8	19.6	18.9	15.5	12.9	10.8	8.6	(22)	
	5%	DB	6.4	6.4	8.9	15.3	19.8	23.3	26.0	23.7	18.7	14.2	10.9	8.6	(23)	
		MCWB	5.3	5.1	5.6	9.5	13.3	15.8	18.8	18.2	14.6	12.4	9.7	7.5	(24)	
	10%	DB	5.4	5.3	7.1	13.0	17.2	20.8	23.7	21.6	17.2	13.2	10.0	7.6	(25)	
		MCWB	4.4	4.2	4.9	8.3	11.8	14.8	17.9	17.0	13.9	11.6	8.9	6.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	8.1	8.8	12.6	17.1	19.5	21.4	21.1	17.7	14.8	12.6	10.0	(27)	
		MCDB	8.3	9.0	13.5	18.2	21.7	24.5	27.2	24.4	21.3	15.8	13.3	10.5	(28)	
	2%	WB	6.5	6.3	7.4	11.4	15.2	17.7	20.4	19.7	16.4	13.6	11.0	8.9	(29)	
		MCDB	7.2	7.1	10.3	16.4	19.6	23.3	26.1	23.8	19.2	14.6	11.8	9.4	(30)	
	5%	WB	5.5	5.3	6.2	10.0	14.1	16.7	19.4	18.7	15.4	12.7	10.0	7.6	(31)	
		MCDB	6.2	6.1	8.2	13.8	18.0	21.6	24.4	22.4	17.7	13.8	10.9	8.4	(32)	
	10%	WB	4.5	4.3	5.4	8.8	12.8	15.6	18.4	17.8	14.6	11.8	8.9	6.5	(33)	
		MCDB	5.3	5.3	6.8	12.3	16.5	19.8	22.5	21.2	16.6	13.1	9.8	7.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	5.7	8.1	10.3	9.8	10.0	9.5	9.3	8.8	7.5	4.9	5.0	(35)	
		MCDBR	5.2	5.0	10.8	15.7	14.5	15.8	15.0	13.2	11.8	7.9	5.0	4.5	(36)	
	5% WB	MCWBR	5.0	4.0	6.9	9.0	7.3	7.6	7.3	6.9	7.1	5.6	4.7	4.4	(37)	
		MCWBR	5.1	4.6	9.2	12.9	11.7	13.3	12.5	11.5	9.8	6.7	4.8	4.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.294	0.325	0.343	0.382	0.377	0.366	0.388	0.386	0.354	0.342	0.313	0.278	(40)	
		taud	2.243	2.307	2.351	2.295	2.380	2.436	2.395	2.400	2.476	2.414	2.298	2.184	(41)	
	Edn at Noon	Edn at Noon	543	703	802	823	856	872	841	814	789	680	514	431	(42)	
		Edh at Noon	47	71	89	110	108	104	106	99	79	64	45	36	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.35	0.92	2.41	3.94	5.23	5.84	5.29	4.15	2.82	1.28	0.46	0.24	(44)	
	RadStd	RadStd	0.07	0.16	0.23	0.50	0.45	0.34	0.61	0.34	0.32	0.20	0.09	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (56 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	-126	-160	N/A	N/A

Nomenclature: See separate page